

CHAPTER 3

TECHNICAL BACKGROUND

This section presents the technologies and/or requirements that were used in designing and developing an online enrollment system at Lyceum of Tuao.

3.1 System Requirements

3.1.1 Hardware

The hardware resources necessary for Online Enrolment System are presented below:

Minimum requirements	Purpose
The Main Serve Computer (preferably with Intel Pentium Dual Core Processor) equipped with • 500 GB Hard Drive • Windows 7,8,8.1,10 and 11 • 4GB RAM	• The back-end user is recommended to use • The suggested hardware specification and • Operating system to obtain system efficiency and reliability.

The system is intended for beginners and professional users. Its design and uses suit the need of the users as well to the staff of the school registrar.

3.1.2 Software

Development Tools	Purpose
Application and Software • Sublime Text Editor 3.0	• It is a popular code editor known for its clean user interface and extensive customization options. It supports syntax highlighting for many programming languages, including PHP,

• MySQL XAMPP • PHP scripting language • Browser	making it a great choice for web development. Sublime Text also offers features like code folding, multiple selections, and a powerful command palette to boost your coding efficiency. • It is a free and open-source software package that bundles several essential components for web development in a single package. It includes: • Apache HTTP Server: This is the web server software that processes incoming requests from users' browsers and delivers web content. • MySQL database management system: This allows you to create and manage relational databases to store data for your web applications. • This is a general-purpose scripting language widely used for creating dynamic web content. PHP code is embedded within HTML pages and runs on the web server before the response is sent to the user's browser. • A browser is a software application used to access and view websites on the internet. It retrieves web pages from a server using the HTTP or HTTPS protocols and renders the content for users to interact with. Popular browsers include Google
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	Chrome, Mozilla Firefox, and Microsoft Edge.
Operating System Platform • Windows 7,8,8.1,10, and 11	• This software is used as the operating system in the implementation of the developed system.
Tools for Documentation	• This software is used for the presentation and preparation of the Online Enrolment System.

3.1.3 Peopleware

Software Developers	Job Responsibilities
• Project manager • System Analyst • Programmer	• Supervise and monitor the entire project activities and its development. • Responsible For researching, planning, and recommending software and system choices to meet an organization's business requirements • Creates the source codes for the development of the proposed system. Must be an expert with the programing language to be used in the development system.

3.2 System Architecture

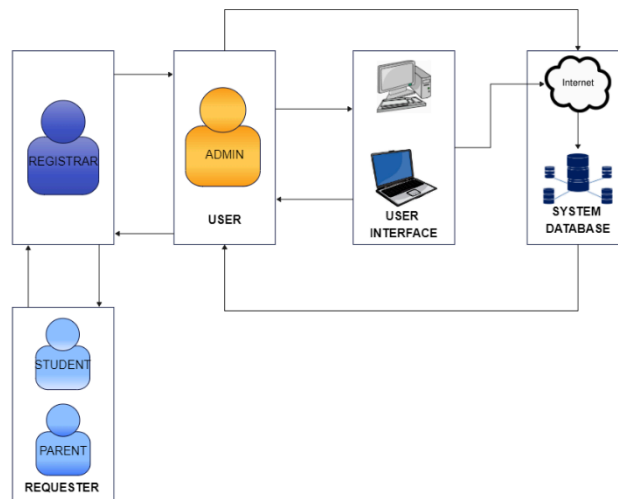


Figure 2. Architectural Design

An online enrolment system typically consists of several key components working together to provide a seamless user experience. At the core, there's a web-based user interface where students can register, login, and manage their enrolment details. This interface communicates with a backend server that handles business logic, such as validating user inputs, processing enrolment requests, and managing session data. The backend server interacts with a database that stores all relevant information, including student records, course details, and enrolment statuses. Security measures, such as encryption and authentication protocols, are implemented to protect sensitive data and ensure only authorized users can access the system.

Additionally, the system architecture often includes APIs that allow integration with other educational platforms and services, such as payment gateways for fee processing or learning management systems for course content delivery. A notification service might be integrated to send alerts and updates to students via email or SMS. The architecture is designed to be scalable, ensuring it can handle a large number of users and transactions, especially during peak enrolment periods. By leveraging cloud services, the system can dynamically allocate resources to maintain performance and reliability.